

Assignment 1 – Starting with Sequential

Fall 2026

Due: Friday, Sep 4 11:59 pm

To-do

Download the zipped folder “sequential” from D2L under contents and complete the following tasks.

Coding Task 1: Complete the sumArray function for calculating the sum of all the array elements.

Coding Task 2: Complete the matrixMultiply function for multiplying two matrices A and B and storing the result in matrix C.

Coding Task 3: Complete the solveQueensUtil function for finding possible solutions for the N queens problem.

Coding Task 4: Complete the bubbleSort and mergeSort functions for sorting values.

Coding Task 5: Complete the numericalIntegration function to approximate the definite integral of a function.

Task 6: Time all five programs (array sum, matrix multiplication, N queens, bubbleSort/mergeSort, and Numerical Integration). Test them using large input values!

Task 7: Post a discussion in D2L explaining which parts of the five programs could realistically be parallelized and how you would split the tasks for all five programs among different processes. Are there any dependencies that force some parts to remain sequential? Which programs are easy, and which one are hard to parallelize. (No right or wrong answer here, simply explain your reasoning and we will parallelize them later 😊)

Deliverables:

- A zipped folder named A1 submitted to D2L dropbox that contains:
 - o Code files for all three programs.
 - o A pdf with screenshots showing the code works and a table documenting the time taken.
 - o A README.txt file
- Available on Github inside the folder where you added me as a collaborator.
- Discussion posted to D2L (You will not be able to see other posts until you publish your own post)

Total Points (70)

- Code runs and works as expected – Task 1: 10 points
- Code runs and works as expected – Task 2: 10 points
- Code runs and works as expected – Task 3: 10 points
- Code runs and works as expected – Task 4: 10 points
- Code runs and works as expected – Task 5: 10 points
- Code timing and screenshots provided – Task 6: 5 points
- Discussion posted – Task 7: 10 points
- Available on GitHub: 5 points